

Citation for published version

Martínez-Carrascal, J.A. [Juan Antonio], Muñoz-Gama, J. [Jorge] & Sancho Vinuesa, T. [Teresa] (2024). Evaluation of Recommended Learning Paths using Process Mining and Log Skeletons: Conceptualization and Insight into an Online Mathematics Course. IEEE Transactions on Learning Technologies, 17(null), 555-568. doi: 10.1109/TLT.2023.3298035

DOI

<http://doi.org/10.1109/TLT.2023.3298035>

Handle

<http://hdl.handle.net/10609/152719>

Document Version

This is the Accepted Manuscript version.

The version published on the UOC's O2 Repository may differ from the final published version.

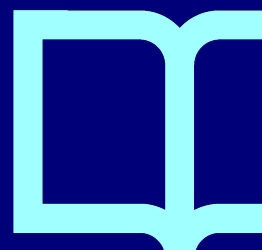
Copyright and Reuse

This manuscript version is made available under the terms of the Creative Commons Attribution license (CC BY)

<http://creativecommons.org/licenses/by/4.0/>, which allows others to reuse it for any purpose, as long as they credit you for the original work.

Enquiries

If you believe this document infringes copyright, please contact the UOC's O2 Repository administrators: repositori@uoc.edu



Evaluation of Recommended Learning Paths using Process Mining and Log Skeletons: Conceptualization and Insight into an Online Mathematics Course

Juan Antonio Martínez-Carrascal, Jorge Munoz-Gama, and Teresa Sancho-Vinuesa

Abstract— Academic institutions dedicate a substantial effort to ensure the academic success of their students. At the course level, teachers recommend learning paths for students to guarantee the achievement of their learning outcomes. In terms of performance, these kinds of approaches are deemed more effective than others based uniquely on performing a collection of independent activities. However, there is no systematic means to validate if following the learning path is effective, nor to assess whether and to what extent students adhere to these recommendations. This paper introduces a novel technique for modelling recommended learning paths, including not only an evaluation of path utility, but also a quantitative measure of student adherence thereto using process mining, and more precisely, log skeletons. Following an event abstraction process regarding real student-recorded activity, a scoping process is employed to retain the trajectories that adhere to the prescribed learning path. The method based on process mining is translated into practice by considering an online university mathematics course. Results confirm the applicability of the method and, in this case, reveal that adhering to the suggested path correlates positively with final grades. Few students strictly follow the prescribed learning path, although the vast majority support it. The method can be easily applied to overcome several challenges associated with enhancing academic performance from the learning analytics perspective.

Index Terms— Academic performance, educational process mining, learning analytics, learning path, log skeleton, student activity modelling

I. INTRODUCTION

THE role of the teacher and instructor in education has always been twofold: to create knowledge by means of lessons and resources, and to recommend the most effective learning strategies [1]. Teachers have traditionally presented topics, such as convenient study sequences and pathways through different learning materials, or convenient timeframes in which to complete them [2]. Today, with the omnipresence of the internet and e-learning management systems, the teacher

role as facilitator has increased significantly, as too has the availability of data generated that can be used to recommend learning strategies [3].

In this context, particularly in online learning, providing recommendations on how to study a particular subject via a *learning path* has become essential to aid students to achieve their objectives. Learning paths (LPs) are defined as sequences of learning activities that should be completed in a particular order to achieve a learning goal [4]. Although providing *recommended learning paths* (RLPs) has been studied in the literature [5]–[10], it remains a challenging and incomplete task [11].

The first problem identified concerns learning variability. The large volume of information and content, the inherent nature of hypermedia, and the intrinsic nature of humans make it almost impossible for two learners to follow the exact same LP, even when learners are required to follow a fixed RLP. Learners that achieve the learning goals presented may progress through the course in their own unique way. Thus, it is necessary to define RLP *models* flexible enough to capture this variability, while also providing sufficient guidance to serve as an RLP [12]. However, the use of models and the identification of sources with which to generate such models is a challenge in itself. Originally, theoretical recommendations were the basis for generating RLPs, due to the complexity of learning processes. Now, a new line of research bases its generation on real learning management system (LMS) trace data on students that have previously fulfilled the goals that educational institutions want future students to achieve.

The second problem identified is the need to establish mechanisms to determine whether and to what extent learners are effectively following RLP models [11]. This element is particularly challenging although it can be applied at both course and student level. At course level, these mechanisms are necessary to evaluate RLPs, particularly in terms of their

convenience for achieving learning goals. Simple binary metrics (followed and not followed) are not detailed enough, and fine-grained metrics (e.g., ranging from 0% to 100%) are necessary to assess RLP relevance and impact on outcomes. At student level, and assuming a favourable proven RLP in terms of academic goals, the detection of relevant deviations between student execution and the RLP can act as warning signs from which to redirect student behaviour.

The present work proposes the use of process mining to address these two problems. Process mining [13] has emerged as a discipline between data science and process science and is composed of a set of techniques and algorithms to extract (or mine) valuable process-related information from event data (known as *event logs*). Unlike other data science techniques that extract abstract patterns in the form of rules or decision trees, process mining creates complete end-to-end process models that capture the different variants of how a process has been executed in a precise but general way. It has been successfully applied to a range of interdisciplinary fields, such as healthcare, finance and education [14]–[16]. Although the use of process mining to analyze LPs is not new [17], it remains nascent, as shown by its proposal of out-of-date approaches (e.g., Alpha algorithms [18]) or the use of model notations (e.g., Petri nets [19]) that are difficult to interpret for non-process mining experts. As a novel contribution, this paper presents the use of *log skeletons*. In contrast to most workflow net-based techniques, the process models resulting from the skeletonization are not only based on the directly-follows relation, but also provide circumstantial information on the event log rather than merely sequential information [20], [21].

This work contributes a novel approach by providing a process mining-based methodology for generating and evaluating RLPs. The proposed methodology consists of the following steps: 1) *abstract an event log*, the necessary input for process mining algorithms, from raw data generated by the LMS; 2) *scope the event log* and select representative traces of the RLPs; 3) generate *log skeletons* as modelling notation to capture RLPs and the required approach to *discover* the model from the previously selected representative traces; and 4) run *conformance* metrics to measure the similarity between the RLP model and the LP followed by a learner on a 0 (completely different) to 1 (exact match) scale.

Besides this novel methodology, the present paper also introduces the practical application thereof in an online mathematics course offered at a European university. The case study includes 330 students and raw data was obtained from the university Moodle LMS. Eight representative traces were selected for the generation of the log skeleton model based on strict RLP compliance. From a student perspective, the course results indicate that, while most students broadly follow the RLP (over 50% of students show a conformance value of 0.9 and above), few students strictly adhere to the path (8 out of

330). From a course-design perspective, there is a correlation ($R=0.894$) between following the RLP and academic performance in terms of final grades. The application of the method enables the appropriateness of the RLP at the course level to be assessed, in addition to quantifying individual deviations at the student level.

The remainder of this paper is structured as follows: Section II outlines the theoretical framework and presents the research questions. Section III proposes the methodology in detail and introduces the course used for practical implementation. Section IV documents the results. Section V includes a discussion of the methodology and its limitations. Finally, Section VI concludes and poses possible future lines of research derived from these findings.

II. THEORETICAL FRAMEWORK

In order to obtain an in-depth understanding of RLP modelling and evaluation, this section analyzes the theoretical foundation directly related to LPs (Section II.A). Subsequently, based on the idea of LP as a process, the intersection between process mining and education, which forms the scope of this study, is discussed (Section II.B). The three research questions addressed in this work are then introduced (Section II.C).

A. Learning Paths

At course level, learning paths are defined as “a sequence of learning tasks or activities which are designated to assist the student in improving their knowledge or skill in the particular subject” [4]. Subsequently, several works have analyzed LPs and their personalization [11]. Their implementation requires the organization of learning activities and a definition of how to evaluate student learning, and should be designed to match learning outcome requirements.

According to this definition, two tasks must be accomplished. First, establish a clear sequence of activities; and second, determine that the path effectively helps to achieve knowledge. These tasks demonstrate complexity [4]. The first aspect is linked to the design of the path itself and research shows different approaches for LP design and potential personalization, e.g., [22]–[25]. [22] provides specific resources to learners based on their personal characteristics. In [23], learning objects are sequenced based on learning assessments. [24] considers feedback from the learning to self-adjust learning components and paths. The approach in [25] begins by generating all possible learning paths that match the learning goal, and subsequently selects one based on the use of a decision model.

However, as [11] indicates “evaluation is always one of the main challenging phases in the learning path personalization method”. While this is particularly true for path personalization, its evaluation is required in environments where single paths are suggested for students taking the course.

The evaluation of RLP involves the need to determine the extent to which students are effectively following the path. Otherwise, it would be impossible to ascertain whether the RLP is assisting students to improve their knowledge. Distinct investigations have analyzed how this “sequence of tasks” is effectively materialized [26]. Initial works on path visualization performed a Gantt chart as a simple tool to represent the path [27]. The work in [28] suggests the use of approaches linked to process modelling and business processes and, particularly, the use of business process modeling and notation (BPMN) models. Recent work includes the use of graphs [26], [29] and sequences [30].

While these studies can help to visualize a theoretical learning path, none provides formal methods aimed to compute the extent to which a student is following the path. References to mathematical formulations that address LP evaluations are difficult to find. Yang et al. [4] recommends an LP model for its development, although it omits the evaluation phase. The evaluation is conducted via a specified survey, which revealed that 90% of respondents think it beneficial to link learning tasks with learning results. According to the same study, the model itself provides a clear insight into how students progress through the course.

By working with personalized paths, [31] proposes an evaluation based on the quantification of precision and sensibility. These parameters are linked to the predicted student learning style used for classification, rather than on the path itself. The work in [32] compares results of a control group to one where the personalized learning path was introduced. The example in [33] indicates the importance of analyzing student satisfaction to evaluate RLPs.

Therefore, most research focuses on building and offering tailored RLPs rather than providing formal methods to evaluate these paths. Questions such as: *To what extent is the student following the path?* or *How does deviating from the path affect academic performance?* are not formally addressed. These questions become even more significant in contexts where an RLP is offered, but in which the student is free to choose learning objects.

B. Process Mining in Education

Data collection has always been a key element in learning analytics, with approaches such as questionnaires or think-aloud methods representing some of the most frequently deployed techniques. However, the widespread use of learning management systems (LMS), massive open online course (MOOC) platforms, and open-ended learning environments (OELE) has opened the door to a new type of digital data: trace data [34], i.e., digital footprints of learner interactions with the system. Today, trace data is commonly used to study dimensions, such as self-regulation patterns [34] or the effectiveness of blended courses [35].

Early works dealing with this topic consider discrete

parameters, such as total number of discussion messages posted or mail messages sent [36]. However, as [37] remarks, quantitative measures of activities tightened to LMS parameters often lack meaningful information linked to didactics. Values, such as click amounts or number of days a student logs in, may reveal information about how often and to what extent they perform activities, but not about what he/she is actually doing. It also lacks the temporal perspective, and nothing can be inferred regarding potential previous content not covered or a lack of previous activities. That is why an increasing number of recent works aim to detect patterns emphasizing the role of temporality and sequences of events, in contrast to more conventional statistical analysis [34]. Accordingly, this is the approach adopted in the present work.

Naturally, this focus on the sequences has been the perfect application domain for techniques such as transition and sequence analysis [37], [38], and more recently, process mining [15], [39]. Process mining combines process science and data science to extract meaningful insights from trace data, in order to enhance understanding and improve outgoing processes [13], whereby LPs are understood as processes in this work. As reported by [40], process mining provides useful tools for analysis while sequence analysis provides enhanced predictive results.

The most used family of process mining techniques are those with which users can discover (or generate) an end-to-end process model, for example, the discovery techniques of alpha miner and inductive miner [13]. In the approach followed in this investigation, the authors use trace data from certain exemplary students to model and evaluate a recommended LP.

Different approaches can be used to select representative traces, depending on the intended effect of the RLP. These include more straightforward grades or pass/fail approaches and the use of more sophisticated criteria, such as desired learning style or student wellbeing [11]. These traces are translated into a model using distinct techniques, such as clustering, association rule mining, Markov models or collaborative filtering. Clustering can be accomplished with techniques such as k-means or hierarchical clustering, which recommend a path based on the most frequent ones taken by learners in a given cluster. Association rule mining entails identifying frequently occurring sequences of learning activities and employing them to recommend learning paths. Markov models involve constructing a transition matrix based on the learning traces and using this to predict the most probable sequence of activities for a particular learner. Collaborative filtering involves identifying learners who have completed comparable activities and recommending activities that the current learner has not yet completed. For a comprehensive overview of these methods and their application in the literature, readers should see [11].

Furthermore, direct selection or the definition of general (possibly theory-driven) scoping rules that all selected

representatives must fulfil are also possibilities. These options have the advantage of simplicity.

The literature of process mining techniques for education of is wide [37], [38]. Although most techniques are similar regarding their goal, the approach and notations used differ. For example, similar to our work, a process model notation such as Petri nets, which is widely used in process mining, has also been used to find LPs or to optimize the path that a student must follow to ensure a successful outcome [41]. However, in a similar way to other process model notations, including BPMN, such process notations have a limited reach in terms of learning analytics due to their procedural nature and formal semantics, despite their broad use in the context of business processes. In contrast, it is more common in learning analytics to use discovery techniques that result in heuristic nets or directly-follows graphs (DFG), due to the fact that they are easier to understand for non-process mining experts. In [42], and extended further in [43], the authors propose the use of the DFG technique and its implementation in the tool Disco [44] to analyze student LPs during a learning session. In [45], a similar approach is utilized, in this case with the implementation of DFG in the tool Celonis [46] to analyze the LPs during emergency remote teaching implemented as a result of the COVID-19 pandemic. Although easy to generate and understand, DFG and other similar process notations also present their own particular problems. These include a lack of explicit elements to model concurrency, misinterpretations when interactive filtering is applied, or only incipient efforts to establish a semantics, among others [47], [48].

Different approaches can be used to obtain a process model, such as the seminal ones described in [49] in which the range of techniques used is broad. Relevant examples include the alpha algorithm (including extensions of the original), heuristic miner, genetic miner and decision tree miner. An initial overview of methods can be found in [50] and [51]. The generated models can be expressed in different notations, with Petri nets the predominant form [51]. This notation can be translated to other representations, such as BPMN, DFGs, BPEL (business process execution language) or UML (unified modeling language) activity diagrams, although resulting models may not be entirely equivalent and differences in terms of accuracy or simplicity may arise in the conversion [52].

An alternative approach to process discovery, more closely related to the one presented in our work, is the use of declarative process models rather than procedural languages [53]. In such notations, process models are built from a set of relations extracted from the trace data. While procedural languages may be used for predictable environments, techniques based on declarative languages can be useful in cases which involve high degrees of variability, such as the path followed by different students [53].

The work described in the present study is founded on log

skeletons [54], which is a recent approach adopted in relation to procedural process mining, and one that has proven to be more accurate than alternative discovery techniques [54]. The process skeletonization presented in [20], [21] combines more traditional relations, including an action directly following another action in a learning path, with new relations, such as the equivalence of two actions based on their occurrence within a learning path, or the counting of minimal and maximal occurrences of an action in a learning path [54]. It is of equal importance to ensure that current implementations of log skeletons, such as the one in ProM [55], allow for an interactive explication of discovered process models, via the selection of one or several relations. In the approach detailed herein, log skeletons are discovered from exemplary learning paths to create an RLP that can be explored and transformed interactively.

The second most used family of process mining techniques are those that utilize trace data to measure the degree to which the execution of a process conforms to a given process model. Conformance checking has always been explored to a lesser extent in process mining and, thus, its application in learning analytics has followed a similar pattern. In [56], behavioural chain conformance is used to study the impact of following or not following the designed LPs. A different approach is studied in [57], in which the authors introduce the advantages and disadvantages of using Petri-net based conformance checking to identify deviations between students and their recommended study plans. For log skeletons, conformance can be quantified based on the number of relations that fail to match between an RLP and the learning path of a particular student. However, as far as is understood by the authors of the present study, there has been no prior use of skeleton-based conformance checking in educational scenarios. Accordingly, the approach followed in this study is the first time researchers have explored the use of skeleton-based conformance checking to numerically assess the degree of adherence between a student LP and the RLP, without reliance placed on rigid procedural notations, such as Petri nets.

Finally, certain works propose the combination of process mining techniques with additional information science techniques, to improve understanding with regards to LPs. For example, the application of clustering techniques prior to process mining in [58] in order to enhance the fitness and comprehensibility of the generated process models. A further example is [59] in which process mining is used to create training data for the application of gradient boosting decision tree techniques to measure the improvement of the problem-solving abilities of students. In [60], the authors combine stochastic analysis and temporal event data in terms of first-order Markov models. The combination of techniques results in a highly informative merging of both and, therefore, the present study could be also positively impacted by combining it with other approaches.

C. Research Questions

By linking the aforementioned ideas, the potential of process mining, particularly process skeletonization, is explored to model and evaluate RLPs. This gives rise to the following research question that addresses the theoretical foundations behind the process:

RQ1: How can process skeletonization be used to model RLPs based on LMS activity traces?

The practical implementation of the methodology is also assessed. Considering the process mining approach adopted in this study, in addition to its focus on RLPs, two additional research questions based on a real case study are posed:

RQ2: For the case under analysis, and based on the model generated, to what extent do students follow the RLP? Is this follow-up linked to course grades?

RQ3: Considering academic activity, what differences can be found between students who strictly adhere to the path and those who do not?

In terms of process mining, RQ2 links to conformance while RQ3 envisions process enhancement. Both serve as validation for the proposed methodology, which should be applicable to any course in which RLP evaluation is under consideration.

III. METHODOLOGY

This section provides an overview of the proposed methodology to address the research questions introduced in Section II.C. Section III.A assesses the four steps of the methodology. This methodology will be applied to a specific case-study, whose context is presented in Section III.B. Subsequently, Section III.C details each one of the four steps in the methodology. Finally, Section III.E. covers the data preparation process, including those details linked to the technical implementation.

A. Method Overview

An overview of the proposed methodology is shown in Fig. 1. The methodology uses the data generated by the LMS to create and evaluate an RLP using process mining techniques. Therefore, the first stage of the methodology is to shape this raw data into an event log structure, the necessary input for process mining algorithms. By means of a specific event abstraction, the resulting event log is translated into a sequence of meaningful activities that reflect all the LPs followed by each student in the course. In contrast to other approaches, this proposed method relies on the use of desired (or representative) examples of LP traces to generate the RLP. Thus, in the second stage, certain LPs in the event log are scoped and selected. In

the third stage, these representative traces are used to automatically generate (or discover) a process model that generalizes the RLP, using process discovery techniques. It should be noted that this methodology is the first to utilize process skeletonization as a modelling notation for RLP. In the final stage, approaches from another process mining technique, known as conformance checking, are used to numerically evaluate the extent to which a given LP follows or does not follow the discovered RLP.

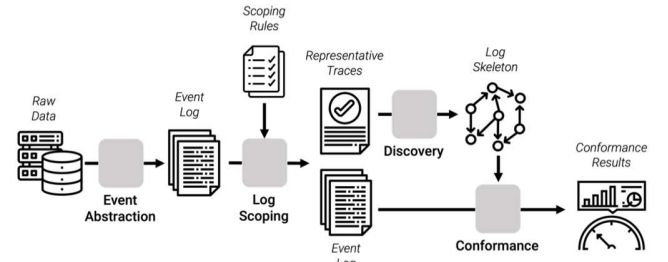


Fig. 1. Overview of the proposed methodology, including its four steps, and all the inputs and outputs involved.

B. Context

To evaluate the feasibility of the proposed methodology, the present study illustrates its implementation in a real course: an introductory mathematics course offered by an online university in Europe. It is a one-semester course taught to all students upon entry to engineering programs. Enrolled students engage in asynchronous online activities through a Moodle-based LMS. The course contains a sequence of 11 modules that feature training and evaluation exercises. Due to its higher complexity, Modules 10 and 11 are split into two parts (A and B). A non-graded, introductory quiz explains how the course is structured and assesses prior knowledge.

A recommendation is presented to students by the instructors at the beginning of the course. Successive iterations of the course have emphasized retention of students by providing a clear sequence of recommended activities. Every week constitutes a new module (except for Modules 10 and 11, which are spread over two weeks each), and an evaluative assessment is conducted. There are two recap tests, one after Module 5, and another following Module 11. In addition to these evaluative assessments, students are encouraged to complete weekly practice tests. These tests can be taken at any time and multiple times, as necessary. In the event of retaking a test, the questions are changed in order to aid student learning, rather than simply repeating the same test. Students are encouraged to carry out these tests as many times as necessary to improve their understanding of the key concepts. Students receive immediate feedback following the completion of these practice tests. Once they feel sufficiently confident, each student should take the evaluative assessment for the module.

Crucially, modules should be completed in sequence. To

ensure that students are motivated to follow the RLP, modules are not accessible until the indicated start date. Furthermore, non-evaluation assessments are administered prior to those that impact on grades, to help promote student practice. Regarding evaluative activities, deadlines are provided prior to the start of the subsequent unit. No submissions are accepted after the deadline. Once completed, practice and evaluative activities are available throughout the course and can be utilized by students for review.

Regarding the activities impacting grades, thirteen assessment activities linked to the different modules are performed throughout the course, with one associated to each module, except modules 10 and 11, that are assessed through two evaluations each. Two additional recap tests are also performed (one after Module 5, the other one at the end of the course). For the purpose of calculating student grades, assessment scores are first grouped into two blocks. The first block corresponds to the first 5 modules; the second spans 6 to 11. For each block, the lowest score is discarded, and the remaining grades are averaged. To pass the course, each of these two computed averages must be at least C-. The average of the block scores and the recap tests must be at least C+, which is the minimum passing grade. With the aforementioned grade, it is not possible to pass the subject, as is the case with a final exam.

It is noteworthy to observe how the RLP takes shape in light of the proposed structure. Ideally, students should engage in non-evaluative activities multiple times before taking the evaluative assessment for each of the modules following the correct order. While this sequence is encouraged, it is interesting to note that deviations from the recommendation are possible. The completion of the questionnaires prior to the evaluation exams is neither mandatory, nor does it impact grades. In addition, the student only needs to pass each block, with the lowest grade of the associated examinations disregarded. Therefore, he/she may even skip the evaluative assessments in specific modules.

With respect to the number of registered students, this study analyzes data associated with six distinct course groups totalling 330 students, of which 280 pass the course. Student responses indicate a high level of satisfaction with the course design and organization.

C. Detailed steps

1) Event Abstraction and Event Log Pre-processing

Since the aim of the methodology is to apply process mining, the first task is to transform the raw data generated by the LMS into an *event log*. An event log contains the *traces* of a process, i.e., an ordered sequence of events that have occurred during the execution of a process instance [52]. In this case, each trace represents a student and the LP followed in the LMS. Event logs can be seen as “footprints” left behind by the execution of the process (i.e., student LPs), and process mining uses these event

logs as a starting point from which to perform distinct tasks. Formally, all traces in an event log must have a unique identifier, which refers to different process instances. Moreover, all events in a trace must be ordered and refer to the *activity* of the process performed. Here, each trace identifies the LP of each student, the activity may represent material read or a quiz performed in the LMS, and the events within a trace are ordered according to the timestamp in which they occurred. An anonymized version of student IDs is consistently used as a trace identifier in research into the modelling of courses as a process [17].

While the direct use of low-level activities (e.g., watch video 4.2, complete quiz 7.5, etc.) for traces is technically possible, it is far from encouraged in the education context, since it is highly likely to result in *spaghetti models*, i.e., models that are based on such a large quantity of activities and relations that they become unusable [61]. Rather, the literature suggests the combination of low-level activities within more abstract and meaningful events, thus reducing complexity and improving understandability [62].

In this case, use was made of the modular structure of the course and was combined with on-time or not on-time access to the content, in order to define the high-level event log, since a student may engage in practice or evaluation activities for each module. Practice is considered genuine if completed by the appropriate deadline. However, the student may prefer to postpone such practice, in which case it would constitute a late activity. The student can also complete the task on time and review it later. In this case, it is considered a review action. These concepts are shown in Table I and can be easily obtained from the raw data in any LMS.

TABLE I
HIGH-LEVEL ACTIVITY DESCRIPTION

High level activity mnemonic	Description
QINIT	Introductory Questionnaire for the course (not mandatory) providing details into the course
Mn-PR	practice (non-evaluation) activity for module <i>n</i>
RECAPx-PR	or for recap test <i>x</i>
Mn-EV	evaluation activity for module <i>n</i> or for recap
RECAPx-EV	test <i>x</i>
Mn-REVIEW	review of content associated with module <i>n</i> or
RECAPx-REVIEW	with recap test <i>x</i> once the module or the is complete. The content has been completed by the due date.
Mn-LATE	work on content associated with module <i>n</i> after module is over. This LATE content has not been completed by the due date

To clearly illustrate this concept, consider a student who performs a non-evaluation quiz associated with the first module. According to course recommendations, performing this exercise while the module is active (the student is practising) is different to performing it later; in the latter, the learner is either late or reviewing the content, depending on whether the material has already been covered. While these

low-level events are recorded in a similar way in the LMS database, they correspond to distinct high-level learning process activities. Once this event abstraction has been performed, more meaningful learning processes are obtained. As an example, the present investigation has performed the event abstraction process on the log records associated with a random student and represented the results using DFG notation. Fig. 2 shows the resulting representation. For clarity, the visualization is limited to the activities performed by the student along the initial modules of the course under analysis.

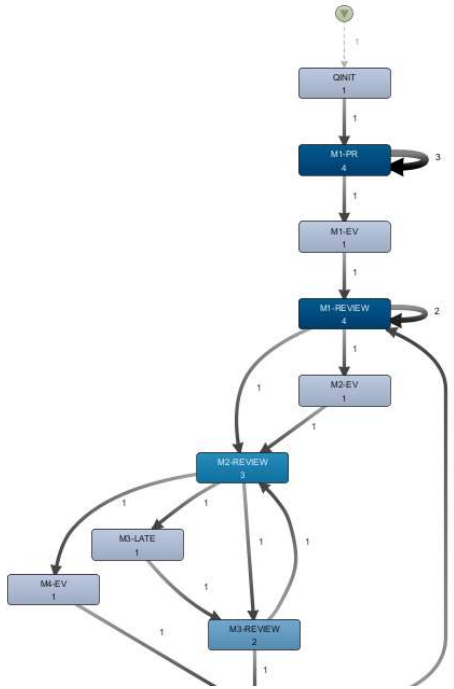


Fig. 2. Representation of the LP followed by a sample student during the initial stages of the course (in DFG model notation). Nodes represent high-level activities. Edges represent transitions among the nodes, including the number of transitions present in the log.

The LP in Fig. 2 is represented using the DFG notation, which is the most used model notation among commercial process mining tools [44], [46]. Although intuitive (i.e., an arc shows how one activity directly follows another), the interpretation of DFG models can be misleading and is beyond the scope of this paper. Interested readers should see [47] for further details. Using the mnemonics in Table I, the nodes in Fig. 2 represent high-level activities. In this figure, the student begins with the introductory course quiz and then attempts non-evaluation quizzes. Following work on these quizzes, the student decides to take the evaluation assessment. He/she subsequently reviews the content in module 1 prior to attempting the evaluation assessment for module 2. This content is reviewed (M2-REVIEW) and he/she then moves on to cover module 3 content. This latter step is undertaken late

and subsequently reviewed. The image also shows the evaluation activity associated with module 4. No reference is found regarding module 3 content being covered. The path taken by the student can be analyzed and, as can be seen, the student in this case skipped content related to module 3 and, therefore, did not follow the LP recommended by the teachers.

2) Log scoping and representative traces

In contrast to other techniques that directly model a (more or less flexible) RLP, this approach explores the use of real execution LPs as representative examples to automatically create a model that generalizes such plans in an accurate manner. In this step, the event log is scoped to select a set of representative traces to use in the subsequent step.

This research uses the definition of the scoping rules based on the recommendation as defined in Section III.A, which translates into three general scoping rules with which to select representative traces, as follows:

- Practice should be undertaken for all modules prior to performing evaluation assessments associated with that module.
- Evaluation assessments must be performed for all modules.
- Students should take part in both recap tests.

These scoping rules derive from the course structure and related considerations are presented in Section III.A. Students are encouraged to engage in the non-grade-bearing activities prior to undertaking evaluation assessments and not to skip any evaluative assessments. However, they may decide not to strictly follow these rules.

The analysis of the log indicates that only eight students strictly verify the three scoping rules defined for the present case study. Their associated traces combine theoretical guidance with real executed LPs. Nevertheless, it should be noted that this does not correspond to a single LP. For example, there are no limitations on content review. Certain students may opt to review (for instance) Module 2 material, while others may not. Some may perform this review after Module 3, others after Module 4.

3) Log Skeleton and Discovery

Our approach to process discovery is based on log skeletons [54]. Rather than attempting to capture all the information in the log and convert it to a process notation, log skeletons focus on and represent the relations found in the log, particularly an equivalence and frequency of activities and activity ordering (analysis of whether certain activities are always found prior or subsequent to others). Table II summarizes the content of the

skeleton.

TABLE II
CONTENT OF A PROCESS SKELETON

Concept	Description
Equivalence	Detection of activities that show the same frequency in all running instances of a process
Always after (A,B)	Detection of activities (A) that are always followed by certain others in the future (B)
Always before (B,A)	Detection of activities (B) that are always preceded by certain others in the past (A)
Never together	Detection of activities that are never found together in a single trace
Directly follows (A,B)	Detection of activities (A) that are always followed by another activity (B) in all traces considered
Activity frequency	Detection of maximum number of times each activity is performed in the traces analyzed

This information is integrated into a skeleton that contains the “information on the structure in the log using a number of relations” [54]. Log skeletons represent an alternative strategy to Petri nets or BPMN models, which are procedural process models that seek to define end-to-end processes in which activities may only be executed when explicitly triggered. Conversely, log skeletons generate a constraint-based model consisting of a set of rules that determine the expected process. This approach has been shown to outperform other automated discovery algorithms in terms of classification, and indeed won the Process Discovery Contests of 2017 and 2019 staged at the International Conference on Process Mining [54]. This classification potential is used herein to determine whether student behaviour can be considered to match that suggested by the instructor.

4) Conformance Checking

In process mining, *conformance checking* refers to a process mining technique that receives an event log and a process model as input and analyzes the extent to which behaviour observed in the log is captured in the model [63]. Conformance checking includes metrics to numerically quantify conformance value, techniques to collect mismatches, and approaches to diagnose conformance discrepancies [64], [65]. Several families of conformance checking algorithms have been proposed in the literature: from footprint and token-replay techniques to more recent alignment-based techniques. While some are modelling-notation specific (e.g., token-replay for workflow nets notation) [66], others are defined independently of notation (e.g., alignment-based conformance checking) [13].

For the purpose of this work, the metrics proposed and implemented to check the conformance value between an event log (student LPs) and a log skeleton model (describing a desired RLP) have been selected consistently with [54]. The implementation of the conformance algorithm measures the degree of deviation of the trace (the student LP) from the recommended behaviour modelled in the log skeleton model. In particular, given the declarative nature of the log skeletons, the conformance score is determined by computing the relations

mismatch between a log skeleton and a student trace. This metric results in a single decimal number between 0 and 1, where 0 represents a total mismatch between a student LP and RLP and 1 represents a perfect match between the two. This facilitates interpretation, a key element of the methodology proposed in this work. Regarding the mathematics course, conformance values were computed for all student LPs and the RLP log skeleton obtained in the previous step.

D. Preparing data

1) Event abstraction

Each Moodle log line indicates the associated component, context and event name. The goal of the event abstraction process is to convert these Moodle-specific data into process activities, such as those in Table I, which constitute the seed of the discovery process.

In this case, and in accordance with the instructional design outlined in section III.B, the present research focuses on the practice and evaluative activities. This requires focusing on the analysis of the Quiz component at the Moodle log level. Table III visualizes the initial event abstraction process carried out on the log. The context of the event determines the module on which the student is working. The specific event provides pertinent information about student activities and schedules. In this case, the events associated with quiz initiation, completion, and evaluation submission are filtered.

TABLE III
OVERVIEW OF THE EVENT ABSTRACTION PROCESS

Date	Context	Component	Event	Related process activity
17/02	Maths 101	System	User enrolled in the course	Non relevant
17/02	Maths 101	System	Course viewed	Non relevant
17/02	Quiz: Initial Quiz	Quiz	Course module viewed	QINIT
17/02	Quiz: Initial Quiz	System	Quiz attempt viewed	QINIT
17/02	Quiz: Initial Quiz	System	Quiz attempt submitted	QINIT
21/02	Numbers: Practice	Quiz	Course module viewed	M1-PR
21/02	Numbers: Practice	Quiz	Quiz attempt started	M1-PR
21/02	Numbers: Practice	Quiz	Quiz attempt viewed	M1-PR
21/02	Numbers: Practice	Quiz	Quiz attempt viewed	M1-PR
21/02	Numbers: Practice	Quiz	Quiz attempt viewed	M1-PR
21/02	Numbers: Practice	Quiz	Quiz attempt submitted	M1-PR
21/02	Maths 101	System	Course viewed	Non relevant

Simplified extract of the log to show the abstraction process. The log shown corresponds to a filtered version of the Moodle log for a given user. Dates shown are limited to day/month for readability.

Likewise, all events associated with a single submission are

analyzed and potentially merged into a single event. To do so, the actual and intended execution times are compared in order to determine whether the activity is timely or late. Similarly, it can be determined whether the content constitutes a review activity by analyzing whether the student has previously conducted a review thereof.

The execution of the abstraction process translates an initial log containing 342,596 lines into 20,698 meaningful events associated to the 330 students enrolled in the course. The sequence of abstracted events associated with each student constitutes a case in the process-mining nomenclature. As can be seen, and despite the simplification, each student performs an average of 61 process activities throughout the course.

2) Technical aspects

The event abstraction and event log creation described in the first step of the methodology were performed on the mathematics course raw Moodle data through Python scripts. The PM4Py library (version 2.7.2) [67] was used to implement the algorithms. In particular, the *discover_log_skeleton* inside the *pm4py.algo.discover* package was used to build the model. The graphical view of the resulting log skeleton can be generated with the log skeleton visualizer plugin for ProM 6.9 [55]. For a detailed description of the plugin internals and the log skeleton interpretation, see [20].

The conformance measure was computed using the aforementioned PM4Py library, by means of the *conformance_log_skeleton* algorithm within the *pm4py.algo.conformance* package.

IV. RESULTS

This section discusses the results of having applied the methodology proposed to the mathematics course case study and provides answers to the research questions posed in Section II.C. To reiterate, this methodology is generalizable and can be applied to any course with logged activity (e.g., LMS), as long as this logging includes action performed, student identifier and moment in time.

A. RQ1: How can Process Skeletonization be used to Model RPLs?

To model the RLP we follow the method outlined in Section II.B. The three scoping rules defined in Section III.C were used to scope the log, resulting in eight selected representative traces. These eight representative traces were used as input for the discovery algorithms in order to generate an RLP log skeleton process model that generalizes all exemplary behaviour in an accurate manner. A view of the skeleton is shown in Fig. 3. For clarity, the skeleton has been filtered to reflect only the activities directly linked to the initial quiz and the those associated with the first two course modules.

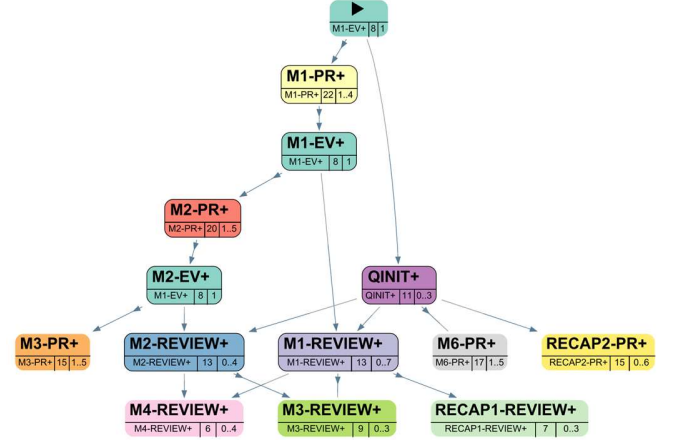


Fig. 3. Resulting skeleton for the mathematics course under analysis. For clarity and readability, only relations linked to activities from the initial quiz and first two modules are reflected.

Despite the low number of traces selected, the skeleton reflects a high degree of variability. The initial stages of the course strictly follow the RLP, although the sequence of activities does not follow a straight path. For instance, Fig.3 shows that some students prefer to start the course by completing the initial questionnaire that explains the course's fundamentals, while others proceed directly to M1-PR. Similarly, after completing the evaluation for M1 (M1-EV), some choose to practise for the second module (M2-PR), while others choose to review the first module's material.

The skeleton in Fig. 3 incorporates the scoping rules in addition to information resulting from the inclusion of real activity. For a detailed description of log skeleton interpretation, see [20]. Related to Fig.3., and besides relationships, activity frequency is well understood. For example, there are 22 instances associated with the practice of module 1 (M1-PR), whereby the activity occurred between once and four times each trace.

B. RQ2: To what extent do Students Follow the RLP? Is this linked to grades?

Once the log skeleton process model representing the RLP was generated, the conformance value of all student LPs, represented as traces in the event log, were measured. Conformance values fall into a $[0,1]$ range, with 0 indicating the lowest degree (0%) of match with the RLP. Results are shown in Fig. 4 (a) related to final course grade and (b), via a histogram.

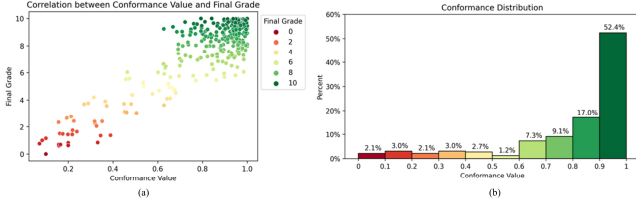


Fig. 4. Correlation between conformance value and final grade (a) and conformance distribution (b).

As Fig. 4 shows, 52.4% of students have conformance values that range from 0.9 to 1, i.e., approximately half follow the RLP perfectly or almost perfectly. Conversely, the percentage of students far from following the RLP (i.e., conformance values between 0 and 0.4) is 10.2%.

The previous analysis shows the extent to which students follow the course RLP and where the differences arise. However, it does not evaluate whether, by following the RLP, students are likely to achieve better academic results. To do so, the conformance value was compared with the final grade of each student. Results are shown in Fig. 4 (a). Results indicate that students with higher RLP conformance also obtain higher grades. Correlation between the conformance value and final grade is $R=0.8942$ and the coefficient of determination is $R^2=0.7996$.

The purpose of this analysis was to illustrate the use of the proposed methodology to generate an RLP from real representative examples, measure the conformance value of student LPs with respect to RLPs, and correlate these values with a desired outcome, such as grades. The discussion of grading as a valid academic performance proxy is beyond the scope of this paper, and each course owner may wish to maximize a different outcome depending on each case (e.g., grades, pass/fail ratio, course satisfaction, time spent, dropout, among others).

C. RQ3: What Differences can be Found?

The results above quantify the similarity between the RLP model, and the LP followed by the students. This was one of the goals pursued. However, the analysis of the skeleton itself and the deviations computed for each trace provide deeper insight linked to behavioural differences.

The declarative nature of the skeleton ensures that an analysis can be undertaken regarding which rules are violated for each case. In other words, the identification of where the process diverges from the expected behaviour contained in the skeleton. By comparing the skeleton with a given trace, it is possible to determine the most common divergences between the RLP (modelled in the skeleton) and a given case (i.e., a real student path).

To determine where the most significant differences lie, emphasis is placed on two aspects: the number of times each action is performed on average compared to the skeleton, and

the analysis of violating transitions. Particularly, analysis of “always before” and “always after” transitions was conducted, since the skeleton in this case study did not capture “never together” relationships. The equivalence relationship between activities does not provide useful details in the present case study.

Beginning with the comparison of the number of times an activity is performed, averages have been quantified and evaluated. Table IV combines the results and describes where differences are more notorious, in addition to the top-5 activities showing higher differences in terms of frequency.

TABLE IV
DIFFERENCES IN THE FREQUENCY OF ACTIVITIES PERFORMED DURING THE COURSE. (EXPRESSED IN %). COMPARISON OF STUDENTS CONFORMING RLP VS. NON-CONFORMING

Activities more frequently performed by non-conforming students		Activities more frequently performed by conforming students	
Activity	Difference (%)	Activity	Difference (%)
M11A-REVIEW	70.88	M9-PR	57.44
M11B-REVIEW	68.19	M7-PR	51.13
M7-REVIEW	63.92	M10A-PR	50.49
M8-REVIEW	63.50	M8-PR	49.85
M4-REVIEW	57.69	M11A-PR	49.38

The results indicate that students who adhere to the RLP conduct greater amounts of prior practice before evaluation, especially during the second course block (starting from Module 6 onwards). In contrast, students who deviate from the path have a greater number of content reviews following the assessment. Interestingly, this increase is most prominent during the second course block.

The violations to the “always after” and “always before” relations contained in the model of the RLP reinforce these findings. Regarding “always after” transitions, they reveal that although students participate in evaluation activities in the early phases, they do not always practice prior to evaluations in later modules. For example, the skeleton indicates that while the student takes the evaluation in an early module, such as M2 (M2-EV), he/she does not take practice tests before sitting the assessment for Module 11B. Similarly, and in an even clearer manner, violations of the “always before” relationships indicate that evaluation is frequently performed without prior practice in later modules (i.e., Module 8 or Module 9). Table V displays the most common violations.

TABLE V
MOST COMMON VIOLATIONS TO RELATIONS PRESENT IN THE SKELETON

Always after relations	Always before relations
(M2-EV, M11B-PR)	(M10A-EV, M9-PR)
(M1-EV, M11B-PR)	(M9-EV, M9-PR)
(M2-EV, M10B-PR)	(M8-EV, M8-PR)
(M1-EV, M10B-PR)	(M5-EV, M5-PR)
(M3-EV, M11B-PR)	(M10A-EV, M8-PR)

Due to the constraints imposed by the scoping rules, there is no late content review among students who follow the path, although there are instances of this occurring among certain traces of other students.

The academic implications of these results are beyond the scope of this paper, as its methodological focus lies on demonstrating that common divergences can be identified, rather than assessing their educational significance. Notably, the declarative character of the skeleton facilitates the identification of common divergences.

V. DISCUSSION

A. RQ1: Modelling RLPs

Section III.A synthesizes the method implemented, whose practical implementation generates the graphical representation in Fig. 1. The process skeletonization is an essential component with which to model RLPs.

It is important to note that no prior references were identified in the literature performing RLP modelling as processes in a methodological manner, which produce quantitative conformance metrics with straight interpretation, and that can lead to evaluate the convenience of RLPs.

Considering that the log skeleton is built based on representative traces that contain actual desired behaviour, the result of the conformance checking reflects the amount to which a trace, i.e., the student LP, conforms to the key RLP characteristics. This fact has remarkable implications. First, from a teaching perspective, it provides information on whether the students are effectively following the RLP, with a clear quantitative view at both the group and individual levels.

The method outlined in the present study converts the recorded activity of a student into an analyzable process, thus determining the correlation between the activity performed and the one suggested by the instructor. This method and its quantitative approach can assist with a variety of learning analytics-related issues and queries. First, it enables LP validation. Second, it demonstrates that following the path leads to a superior learning outcome, a highly valuable discovery. This is particularly significant regarding RLP creation. Third, while this work has concentrated on evaluating a single path, since the course under consideration lacked personalized options, it would be simple to extend this function to compare different RLPs.

The procedure is also advantageous from an intervention perspective. The proposed method can be followed over time to determine whether a student deviates from the path. The objective of this approach was not to pinpoint a precise period in time or divergence threshold at which intervention is justified. Rather, it relies on meaningful data and adds pedagogical procedural information that is absent in other approaches. The use of log skeletons is encouraged to match the analysis of modelled and implemented processes from the data

science perspective. Indeed, the results confirm the potential of the method proposed by [54].

B. RQ2: Measuring adherence to the RLP

Conformance measures obtained for the present case study indicate that 52.4% of students achieve high conformance values (over 0.9). Findings also show that following the path had a clear correlation with passing the course, thus indicating a strong positive linear relationship between the two variables. From a learning design perspective, this permits to validate RLPs, thus revealing the utility of following the path.

The transversality of the performed process enables a comparison to be made of both the process followed and its specific outcome. First, this research provides a broad overview of the importance of activity, particularly online activity, to student achievement. The link between the process-mapped activity and final grade is consistent with previous research, for example, [35], [68]. The process carried out places the learning path, and, in fact, the learning process, at the centre of the analysis. This aligns closely with works that suggest meaningful analysis rather than prediction ratios [69], [70]. While the goal of the present work was not to establish predictions, as Fig. 4 (a) indicates, the potential of the conformance value as a predictor of the final course grade is evident. This adheres to the findings of [54].

For this specific case study, the coefficient of determination of 0.7996 suggests that the linear regression model explains approximately 79.96% of the variation in the dependent variable based on the independent variable. Additionally, the high correlation value, and specifically the absence of outliers can be attributed to the specificities of this case study, and in particular to the frequent and continuous presence of evaluative assessments – whose execution in broad terms is necessary to pass the subject–. Still, and as Section IV.C reveals, students can decide to practice before evaluation or not, or to follow certain optional contents such as the initial questionnaire. These divergences will be analyzed in detail in the next subsection.

It is important to note that this outcome may not be applicable universally across all cases, and to keep in mind that correlation does not imply causation. The relationship between the two variables may not necessarily be linear. Additionally, there may be other factors that influence the dependent variable that are not captured by the model.

Therefore, the results of the analysis should be interpreted with caution and further investigation may be necessary to fully understand the relationship between following the RLP and academic outcomes. The main focus of the present research is to model the RLP and establish conformance thereto based on student activity. However, to ensure the significance of following the RLP in terms of academic outcomes, establishing a control group would be required to ensure not only correlation, but also causation.

Regarding the specific LP analysis and the potential to also

analyze the global learning process in greater depth, the method described herein extends visualization techniques, such as those compiled in [26] and enables differences to be quantified. Furthermore, the use of log skeletons provides an indicator that permits a straightforward interpretation, due to its clear definitions based on $[0,1]$ intervals.

In relation to comparisons of both paths followed by students, whether considered at a group or individual level, and the recommended paths, this research contributes significantly to existing literature. While some authors suggest the use of surveys [4], which can be helpful, these tools are difficult to implement and limited to specific courses. A similar problem arises regarding the need to implement control and reference groups [32]. The present approach avoids this need by focusing on determining the relevance of the path in terms of its relation to the learning outcome. It can also accommodate potential comparisons between different paths. In addition, the approach proposed herein can be considered, regardless of the course. In terms of works that account for parameters such as precision or sensibility [31], the present approach adds a process perspective to classification linked to these measures.

C. RQ3: Analyzing divergences

Besides quantification, the method provides insight into the deviations. Being based on log skeletons, it helps to diagnose diagnoses of where and why deviations occur [13]. Specific examples are found on Table IV and Table V. In general terms, students who strictly adhere to the path undertake more practice, while those who do not conduct reviews more frequently.

Moreover, it was discovered that non-compliant students lack practice prior to evaluation in the final modules of the course. Violations of the skeleton also reveal divergences from a process perspective. Table V shows that students who deviate from the RLP do not always practise for individual modules prior to taking their evaluation. This is especially noticeable in later modules (such as M8, M9, and M10A).

The fact of using log skeletons allows to help diagnoses of where and why deviations occur [13]. Moreover, and similar to other process mining approaches, the proposed methodology could be extended from a forensic analysis to a more impromptu scenario. The use of streaming process mining algorithms [71] could help identify early-stage deviations and trigger a possible intervention with the student. Analysis of these divergences is one of many possible lines of future research.

D. Implications

The proposed methodology has significant implications for the design and development of learning technologies. The technique presented in this work is conceptually clear, replicable, and its interpretive capacity is high. The authors recommend its use in a variety of learning analytic scenarios

due to the transversal relevance of the analysis of LPs, the impact of being able to model them and quantify adherence systematically, and the broad range of envisioned applications. These include the enhancement of the learning process, the development of novel approaches to predictive analysis, and the modelling of student behaviour, among others.

Although the approach presented is exemplified by the grades of the students as a goal, it is straightforward to adapt the methodology to other goals by redefining the scoping rules. The flexible definition they provide makes it possible to determine, for example, which paths certain groups adopt and the resulting effect, or to set more complex objectives, such as the implementation of self-regulated learning (SRL) strategies [72] or the students' course experience [73]. Moreover, the method is applicable to problems related to personalised learning [74].

Notice that the present study focuses on the analysis of RLPs. However, if all student LPs are selected for the creation of the log skeleton, the process model will capture all group behaviour, and questions can be posed regarding how students as a whole have interacted. In other words, it can be used to uncover LPs, even if there is no predefined path. In this context, works such as [75] emphasise pattern recognition and advocate for the unbiased reproducibility of methods such as the one we have presented.

In this sense, it is feasible to compare the LPs undertaken by various groups in order to identify divergences and recommend actions that target specific groups, particularly those from disadvantaged backgrounds. Also, to compare the models associated with students who pass and fail the course, as well as those associated with repeaters and non-repeaters. The detected differences can (and should) lead to a process of continuous course improvement.

The method also provides new opportunities in one of the most explored areas in learning analytics: prediction. Recent research [76] suggests that predictive approaches should go beyond the analysis of specific variables and instead focus on the context and content of the course. This strategy is aligned with the approach proposed in the present study, allowing the authors to go beyond studies that focus on specific LMS indicators. Instead, as presented in [77], it is recommended to seek for measurement instruments that are interpretable and also permit intervention.

In conjunction, the predictive ability opens up interesting possibilities for the design of interventions. The case study presented in this work examines a particular course edition. However, there is no restriction on performing the procedure at various points throughout the course or comparing different editions of the same course. In these contexts, the authors envision the ability to contribute to the definition of alerts, that instructors can use to suggest and redirect students to paths that are more likely to result in student success. Due to the

interpretability of the results, it is also possible to analyze the reasons behind the differences and, consequently, design interventions even at the individual level. This represents a powerful tool for addressing tailored interventions, and it can even be considered to provide automatic feedback to students based on the detection of performance-threatening deviations.

All these considerations demonstrate the versatility, applicability, and repercussions of the proposed contribution, which broadens the use of skeletons and reveals appealing opportunities for the implementation of process mining in learning analytics scenarios. The described results encourage its use as a modelling technique, beyond its demonstrated capacity in terms of pure classification.

VI. CONCLUSION

Process mining has proved to be an important method in the analysis of RLPs. Accordingly, this article has analyzed both the relevance and convenience of following a given RLP and the extent to which students follow that path. The use of skeletons has helped to synthesize the differences between the suggested path, derived from the application of scoping rules on real activity, and the actual path followed by the student via a concise indicator.

The adoption of this technique is encouraged to enhance understanding of the pedagogical implications generated by the method. As noted in the preceding section, analysis can be expanded to identify specific deviations between RLPs and actual executions. This would determine the nature of the difference, in addition to its relevance in quantitative terms. It is particularly interesting in systems that implement personalized LPs. While the example course in this study is built on a conventional method with a clearly prescribed RLP, the comparison of alternative paths may provide valuable insight. Such comparison might be conducted using either systems with personalized LPs or those with A/B testing methodologies. The authors are open to collaborate in works related to these lines of research.

REFERENCES

- [1] R. M. Harden and J. Crosby, "AMEE guide no 20: The good teacher is more than a lecturer - The twelve roles of the teacher," *Med Teach*, vol. 22, no. 4, 2000, doi: 10.1080/014215900409429.
- [2] W. Drexler, "The networked student model for construction of personal learning environments: Balancing teacher control and student autonomy," *Australasian Journal of Educational Technology*, vol. 26, Jan. 2010, doi: 10.14742/ajet.1081.
- [3] E. Baran, A. P. Correia, and A. Thompson, "Transforming online teaching practice: Critical analysis of the literature on the roles and competencies of online teachers," *Distance Education*, vol. 32, no. 3, 2011, doi: 10.1080/01587919.2011.610293.
- [4] F. Yang, F. W. B. Li, and R. W. H. Lau, "An open model for learning path construction," in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 2010, doi: 10.1007/978-3-642-17407-0_33.
- [5] A. H. Nabizadeh, D. Gonçalves, S. Gama, J. Jorge, and H. N. Rafsanjani, "Adaptive learning path recommender approach using auxiliary learning objects," *Comput Educ*, vol. 147, 2020, doi: 10.1016/j.compedu.2019.103777.
- [6] A. Khamparia and B. Pandey, "Knowledge and intelligent computing methods in e-learning," *International Journal of Technology Enhanced Learning*, vol. 7, no. 3, 2015, doi: 10.1504/IJTEL.2015.072810.
- [7] P. Dwivedi, V. Kant, and K. K. Bharadwaj, "Learning path recommendation based on modified variable length genetic algorithm," *Educ Inf Technol (Dordr)*, vol. 23, no. 2, 2018, doi: 10.1007/s10639-017-9637-7.
- [8] M. J. Huang, H. S. Huang, and M. Y. Chen, "Constructing a personalized e-learning system based on genetic algorithm and case-based reasoning approach," *Expert Syst Appl*, vol. 33, no. 3, 2007, doi: 10.1016/j.eswa.2006.05.019.
- [9] C. M. Chen, "Intelligent web-based learning system with personalized learning path guidance," *Comput Educ*, vol. 51, no. 2, 2008, doi: 10.1016/j.compedu.2007.08.004.
- [10] C. L. Bian, D. L. Wang, S. Y. Liu, W. G. Lu, and J. Y. Dong, "Adaptive learning path recommendation based on graph theory and an improved immune algorithm," *KSII Transactions on Internet and Information Systems*, vol. 13, no. 5, 2019, doi: 10.3837/tiis.2019.05.003.
- [11] A. H. Nabizadeh, J. P. Leal, H. N. Rafsanjani, and R. R. Shah, "Learning path personalization and recommendation methods: A survey of the state-of-the-art," *Expert Systems with Applications*, vol. 159, 2020, doi: 10.1016/j.eswa.2020.113596.
- [12] Maciej Serda *et al.*, "Evaluation of the Learning Path Specification," *Educational Technology & Society*, vol. 14, no. 3, pp. 218–230, 2011, doi: 10.2/JQUERY.MIN.JS.
- [13] W. M. P. Van der Aalst, *Process mining: Data science in action*. 2016. doi: 10.1007/978-3-662-49851-4.
- [14] J. Munoz-Gama *et al.*, "Process mining for healthcare: Characteristics and challenges," *J Biomed Inform*, vol. 127, Mar. 2022, doi: 10.1016/J.JBI.2022.103994.
- [15] A. Bogarín, R. Cerezo, and C. Romero, "A survey on educational process mining," *Wiley Interdiscip Rev Data Min Knowl Discov*, vol. 8, no. 1, p. e1230, Jan. 2018, doi: 10.1002/widm.1230.

- [16] C. dos S. Garcia *et al.*, "Process mining techniques and applications – A systematic mapping study," *Expert Systems with Applications*, vol. 133. 2019. doi: 10.1016/j.eswa.2019.05.003.
- [17] P. Mukala, J. Buijs, M. Leemans, and W. van der Aalst, "Exploring Students' Learning Behaviour in MOOCs using Process Mining Techniques," *Computing Conference*, 2015.
- [18] W. van der Aalst, T. Weijters, and L. Maruster, "Workflow mining: Discovering process models from event logs," *IEEE Trans Knowl Data Eng*, vol. 16, no. 9, 2004, doi: 10.1109/TKDE.2004.47.
- [19] J. L. Peterson, "Petri Nets," *ACM Computing Surveys (CSUR)*, vol. 9, no. 3, pp. 223–252, Sep. 1977, doi: 10.1145/356698.356702.
- [20] H. M. W. Verbeek, "The Log Skeleton Visualizer in ProM 6.9," *International Journal on Software Tools for Technology Transfer*, vol. 24, no. 4, 2022, doi: 10.1007/s10009-021-00618-y.
- [21] M. Kudo, A. Ishida, and N. Sato, "Business process discovery by using process skeletonization," in *Proceedings - 2013 International Conference on Signal-Image Technology and Internet-Based Systems, SITIS 2013*, 2013. doi: 10.1109/SITIS.2013.158.
- [22] J. N. Davies, M. Verovko, O. Verovko, and I. Solomakha, "Personalization of e-learning process using ai-powered chatbot integration," in *Advances in Intelligent Systems and Computing*, 2021. doi: 10.1007/978-3-030-58124-4_20.
- [23] C. Limongelli, F. Sciarrone, M. Temperini, and G. Vaste, "Adaptive learning with the LS-Plan system: A field evaluation," *IEEE Transactions on Learning Technologies*, vol. 2, no. 3, 2009, doi: 10.1109/TLT.2009.25.
- [24] H. I. Liu and M. N. Yang, "QoL guaranteed adaptation and personalization in e-learning systems," *IEEE Transactions on Education*, vol. 48, no. 4, 2005, doi: 10.1109/TE.2005.858398.
- [25] P. Karampiperis and D. Sampson, "Adaptive learning resources sequencing in educational hypermedia systems," in *Educational Technology and Society*, 2005.
- [26] D. B. Ramos, I. M. M. Ramos, I. Gasparini, and E. H. T. De Oliveira, "A new learning path model for e-learning systems," *International Journal of Distance Education Technologies*, vol. 19, no. 2, 2021, doi: 10.4018/IJDET.20210401.0a2.
- [27] P. Teutsch and J. F. Bourdet, "How to see training paths in learning management systems?," in *Proceedings - 10th IEEE International Conference on Advanced Learning Technologies, ICALT 2010*, 2010. doi: 10.1109/ICALT.2010.98.
- [28] A. Adesina and D. Molloy, "A business process management based virtual learning environment : Customised learning paths," in *CSEDU 2011 - Proceedings of the 3rd International Conference on Computer Supported Education*, 2011. doi: 10.5220/0003273203650368.
- [29] R. Cerezo, N. Suarez, J. Carlos Núñez, and M. Sánchez-Santillán, "EGraph tool: Graphing the learning process in LMSs," in *ACM International Conference Proceeding Series*, 2014. doi: 10.1145/2567574.2567596.
- [30] T. Saito and Y. Watanobe, "Learning path recommendation system for programming education based on neural networks," *International Journal of Distance Education Technologies*, vol. 18, no. 1, 2020, doi: 10.4018/IJDET.2020010103.
- [31] K. Govindarajan, V. S. Kumar, and Kinshuk, "Dynamic Learning Path Prediction - A Learning Analytics Solution," in *Proceedings - IEEE 8th International Conference on Technology for Education, T4E 2016*, 2017. doi: 10.1109/T4E.2016.047.
- [32] A. Klačnja-Milićević, B. Vesin, M. Ivanović, and Z. Budimac, "E-Learning personalization based on hybrid recommendation strategy and learning style identification," *Comput Educ*, vol. 56, no. 3, 2011, doi: 10.1016/j.compedu.2010.11.001.
- [33] F. Essalmi, L. J. ben Ayed, M. Jemni, Kinshuk, and S. Graf, "A fully personalization strategy of E-learning scenarios," *Comput Human Behav*, vol. 26, no. 4, 2010, doi: 10.1016/j.chb.2009.12.010.
- [34] J. Saint, Y. Fan, D. Gašević, and A. Pardo, "Temporally-focused analytics of self-regulated learning: A systematic review of literature," *Computers and Education: Artificial Intelligence*, vol. 3. 2022. doi: 10.1016/j.caeai.2022.100060.
- [35] R. Conijn, C. Snijders, A. Kleingeld, and U. Matzat, "Predicting student performance from LMS data: A comparison of 17 blended courses using moodle LMS," *IEEE Transactions on Learning Technologies*, 2017, doi: 10.1109/TLT.2016.2616312.
- [36] L. P. Macfadyen and S. Dawson, "Mining LMS data to develop an 'early warning system' for educators: A proof of concept," *Comput Educ*, 2010, doi: 10.1016/j.compedu.2009.09.008.
- [37] N. Mirriahi, D. Liaqat, S. Dawson, and D. Gašević, "Uncovering student learning profiles with a video annotation tool: reflective learning with and without instructional norms," *Educational Technology Research and Development*, vol. 64, no. 6, 2016, doi: 10.1007/s11423-016-9449-2.
- [38] I. Molenaar, C. van Boxtel, and P. Sleegers, "Dynamic Computerized Scaffolding of Metacognitive Activities in Small Groups," 2013. doi: 10.1007/978-1-4419-5546-3_36.
- [39] M. A. Ghazal, O. Ibrahim, and M. A. Salama, "Educational process mining: A systematic literature review," in *Proceedings - 2017 European Conference*

- on Electrical Engineering and Computer Science, *EECS 2017*, Institute of Electrical and Electronics Engineers Inc., Jul. 2018, pp. 198–203. doi: 10.1109/EECS.2017.45.
- [40] G. Deeva, J. De Smedt, P. De Koninck, and J. De Weerd, “Dropout prediction in MOOCs: A comparison between process and sequence mining,” in *Lecture Notes in Business Information Processing*, 2018. doi: 10.1007/978-3-319-74030-0_18.
- [41] R. Campos-Rebelo, A. Costa, and L. Gomes, “Finding learning paths using Petri nets modeling applicable to e-learning platforms,” in *IFIP Advances in Information and Communication Technology*, 2012. doi: 10.1007/978-3-642-28255-3_17.
- [42] L. Hidalgo and J. Munoz-Gama, “Domain-Driven Event Abstraction Framework for Learning Dynamics in MOOCs Sessions,” in *Process Mining Workshops - ICPM 2022 International Workshops, Bozen-Bolzano, Italy, October 23-28, 2022, Revised Selected Papers*, M. Montali, A. Senderovich, and M. Weidlich, Eds., in *Lecture Notes in Business Information Processing*, vol. 468. Springer, 2022, pp. 552–564. doi: 10.1007/978-3-031-27815-0_40.
- [43] L. Hidalgo and J. Munoz-Gama, “A Domain-Driven Framework to Analyze Learning Dynamics in MOOCs through Event Abstraction,” *Applied Sciences (Switzerland)*, vol. 13, no. 5, 2023, doi: 10.3390/app13053039.
- [44] C. W. Günther and A. Rozinat, “Disco: Discover your processes,” in *CEUR Workshop Proceedings*, 2012.
- [45] R. Divya Sri and M. M. Patil, “Study of Learners Behaviour in virtual learning environment using Process Mining,” in *Proceedings of CONECCT 2021: 7th IEEE International Conference on Electronics, Computing and Communication Technologies*, 2021. doi: 10.1109/CONECCT52877.2021.9622641.
- [46] C. Ullrich, T. Lata, and J. Geyer-Klingenberg, “Celonis studio - A low-code development platform for citizen developers,” in *CEUR Workshop Proceedings*, 2021.
- [47] W. M. P. Van Der Aalst, “A practitioner’s guide to process mining: Limitations of the directly-follows graph,” *Procedia Comput Sci*, vol. 164, pp. 321–328, Jan. 2019, doi: 10.1016/J.PROCS.2019.12.189.
- [48] S. J. J. Leemans, E. Poppe, and M. T. Wynn, “Directly follows-based process mining: Exploration & a case study,” in *Proceedings - 2019 International Conference on Process Mining, ICPM 2019*, 2019. doi: 10.1109/ICPM.2019.00015.
- [49] R. Agrawal, D. Gunopulos, and F. Leymann, “Mining process models from workflow logs,” in *International Conference on Extending Database Technology*, 1998, pp. 467–483.
- [50] J. de Weerd, M. de Backer, J. Vanthienen, and B. Baesens, “A multi-dimensional quality assessment of state-of-the-art process discovery algorithms using real-life event logs,” *Inf Syst*, vol. 37, no. 7, 2012, doi: 10.1016/j.is.2012.02.004.
- [51] A. Augusto *et al.*, “Automated Discovery of Process Models from Event Logs: Review and Benchmark,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 4. 2019. doi: 10.1109/TKDE.2018.2841877.
- [52] W. M. P. van der Aalst, “Process mining in the large: A tutorial,” in *Lecture Notes in Business Information Processing*, 2014. doi: 10.1007/978-3-319-05461-2_2.
- [53] F. M. Maggi, “Declarative Process Mining,” in *Encyclopedia of Big Data Technologies*, Cham: Springer, 2019, p. 8.
- [54] H. M. W. Verbeek and R. M. de Carvalho, “Log Skeletons: A Classification Approach to Process Discovery,” Jun. 2018, doi: 10.48550/arxiv.1806.08247.
- [55] H. M. W. Verbeek, J. C. A. M. Buijs, B. F. Van Dongen, and W. M. P. Van Der Aalst, “ProM 6: The process mining toolkit,” in *CEUR Workshop Proceedings*, 2010.
- [56] D. Davis, G. Chen, C. Hauff, and G. J. Houben, “Gauging MOOC learners’ adherence to the designed learning path,” in *Proceedings of the 9th International Conference on Educational Data Mining, EDM 2016*, 2016.
- [57] M. Wagner *et al.*, “A Combined Approach of Process Mining and Rule-Based AI for Study Planning and Monitoring in Higher Education,” *Lecture Notes in Business Information Processing*, vol. 468 LNBIP, pp. 513–525, 2023, doi: 10.1007/978-3-031-27815-0_37.
- [58] A. Bogarín, C. Romero, R. Cerezo, and M. Sánchez-Santillán, “Clustering for improving Educational Process Mining,” 2014, doi: 10.1145/2567574.2567604.
- [59] F. Liu, L. Zhao, J. Zhao, Q. Dai, C. Fan, and J. Shen, “Educational Process Mining for Discovering Students’ Problem-Solving Ability in Computer Programming Education,” *IEEE Transactions on Learning Technologies*, vol. 15, no. 6, pp. 709–719, Dec. 2022, doi: 10.1109/TLT.2022.3216276.
- [60] D. and P. A. Saint John and Gašević, “Detecting Learning Strategies Through Process Mining,” in *Lifelong Technology-Enhanced Learning*, M. and D. H. and E. R. and S. M. Pammer-Schindler Viktoria and Pérez-Sanagustín, Ed., Cham: Springer International Publishing, 2018, pp. 385–398.
- [61] A. Bogarín, R. Cerezo, and C. Romero, “Discovering learning processes using inductive miner: A case study with learning management systems (LMSs) | Descubriendo procesos de aprendizaje aplicando inductive miner: Un estudio de caso en learning management systems (LMSs),” *Psicothema*, vol. 30,

- no. 3, pp. 322–329, 2018, doi: 10.7334/psicothema2018.116.
- [62] S. J. van Zelst, F. Mannhardt, M. de Leoni, and A. Koschmider, “Event abstraction in process mining: literature review and taxonomy,” *Granular Computing*, vol. 6, no. 3, 2021, doi: 10.1007/s41066-020-00226-2.
- [63] W. M. P. van der Aalst, *Process Mining: Discovery, Conformance and Enhancement of Business Processes*, vol. 136, no. 2. 2011.
- [64] J. Munoz-Gama, “Conformance Checking,” in *Encyclopedia of Big Data Technologies*, S. Sakr and A. Y. Zomaya, Eds., Springer, 2019. doi: 10.1007/978-3-319-63962-8_89-1.
- [65] J. Carmona, B. van Dongen, A. Solti, and M. Weidlich, *Conformance checking: Relating processes and models, relating processes and models*. 2018. doi: 10.1007/978-3-319-99414-7.
- [66] W. van der Aalst, A. Adriansyah, and B. van Dongen, “Replaying history on process models for conformance checking and performance analysis,” *Wiley Interdiscip Rev Data Min Knowl Discov*, vol. 2, no. 2, 2012, doi: 10.1002/widm.1045.
- [67] A. Berti, S. J. Van Zelst, W. M. P. Van Der Aalst, and F. Gesellschaf, “Process mining for python (PM4py): Bridging the gap between process- And data science,” in *CEUR Workshop Proceedings*, 2019.
- [68] Á. F. Agudo-Peregrina, S. Iglesias-Pradas, M. Á. Conde-González, and Á. Hernández-García, “Can we predict success from log data in VLEs? Classification of interactions for learning analytics and their relation with performance in VLE-supported F2F and online learning,” *Comput Human Behav*, vol. 31, no. 1, pp. 542–550, Feb. 2014, doi: 10.1016/j.chb.2013.05.031.
- [69] C. Romero, S. Ventura, P. G. Espejo, and C. Hervás, “Data mining algorithms to classify students,” in *Educational Data Mining 2008 - 1st International Conference on Educational Data Mining, Proceedings*, 2008.
- [70] T. Lerche and E. Kiel, “Predicting student achievement in learning management systems by log data analysis,” *Comput Human Behav*, vol. 89, pp. 367–372, Dec. 2018, doi: 10.1016/J.CHB.2018.06.015.
- [71] A. Burattin, “Streaming Process Mining,” in *Process Mining Handbook*, W. M. P. van der Aalst and J. Carmona, Eds., in Lecture Notes in Business Information Processing, vol. 448. Springer, 2022, pp. 349–372. doi: 10.1007/978-3-031-08848-3_11.
- [72] J. J. M. Mahauad, M. Pérez-Sanagustín, R. F. Kizilcec, N. Morales, and J. Munoz-Gama, “Mining theory-based patterns from Big data: Identifying self-regulated learning strategies in Massive Open Online Courses,” *Comput. Hum. Behav.*, vol. 80, pp. 179–196, 2018, doi: 10.1016/j.chb.2017.11.011.
- [73] K. L. Wilson, A. Lizzio, and P. Ramsden, “The development, validation and application of the Course Experience Questionnaire,” *Studies in Higher Education*, vol. 22, no. 1, pp. 33–53, 1997, doi: 10.1080/03075079712331381121.
- [74] M. Bannert, P. Reimann, and C. Sonnenberg, “Process mining techniques for analysing patterns and strategies in students’ self-regulated learning,” *Metacogn Learn*, vol. 9, no. 2, pp. 161–185, 2014, doi: 10.1007/s11409-013-9107-6.
- [75] J. Jovanović, M. Saqr, S. Joksimović, and D. Gašević, “Students matter the most in learning analytics: The effects of internal and instructional conditions in predicting academic success,” *Comput Educ*, vol. 172, 2021, doi: 10.1016/j.compedu.2021.104251.
- [76] E. Araka, E. Maina, R. Gitonga, and R. Oboko, “Research trends in measurement and intervention tools for self-regulated learning for e-learning environments—systematic review (2008–2018),” *Res Pract Technol Enhanc Learn*, vol. 15, no. 1, pp. 1–21, Dec. 2020, doi: 10.1186/S41039-020-00129-5/TABLES/3.
- [77] R. Conijn, A. Van den Beemt, and P. Cuijpers, “Predicting student performance in a blended MOOC,” *J Comput Assist Learn*, 2018, doi: 10.1111/jcal.12270.
- [78] R. Umer, T. Susnjak, A. Mathrani, and S. Suriadi, “On predicting academic performance with process mining in learning analytics,” *Journal of Research in Innovative Teaching & Learning*, vol. 10, no. 2, 2017, doi: 10.1108/jrit-09-2017-0022.



Juan Antonio Martínez-Carrascal was born in Barcelona, Spain. He received his BsC in Telecommunications Engineering from the Universitat Ramon Llull (URL, 1994) and his MsC in Electronics (URL, 1996). He has a Master in Business Administration from the Universitat Pompeu Fabra (2003), and followed Ph. D. studies in Electronics – Signal Processing and Information Theory (2000) at URL.

He carries out his research within the Learning Analytics for Innovation and Knowledge Application in Higher Education (LAIKA) Group. He focuses on the impact of technology in education and, in particular, on the influence of students’ characteristics and their activity regarding academic performance. He is also an associate professor at the Universitat Autònoma de Barcelona (UAB, Barcelona, Spain).



Jorge Munoz-Gama was born in Barcelona, Spain. Jorge received his M.Sc. and PhD in Computer Science at the Universitat Politècnica de Catalunya. He is Associate Professor and Deputy Director of the Department of Computer Science at the Pontificia Universidad Católica de Chile, and Director of the Human and Process Research Lab (HAPLAB).

He is the author of the book “Conformance Checking and Diagnosis in Process Mining” (Springer 2016). He is an IEEE Task Force on Process Mining Steering Committee member, senior PC member of the BPM and ICPM conferences, and organizer of the “Education meets Process Mining” workshop. His research interests include Process Mining, especially applied to Education, and Medical Education.



Teresa Sancho-Vinuesa is full professor at Universitat Oberta de Catalunya (UOC) in Barcelona, Spain, where she teaches mathematics for engineers and conducts research on e-assessment, feedback and learning analytics as head of the LAIKA Group. Currently, she is the University Ombudswoman. She was a visiting

professor at the Open University UK (2015), University of Southampton (2018) and Cardiff University (2019).

She has been involved in several research and innovation projects concerning the Internet and Higher Education. She is currently concentrating her research efforts on the use of learning analytics for the improvement of online education and learning, and, more particularly, in the evaluation and feedback processes related to STEM studies. She has participated in over 20 technical program committees, has been reviewer in several academic journals and has authored over 75 academic journal and conference papers, as well as writing two books and several book chapters.